EXAMPLE_PROJECTS_ANALYSIS

Comprehensive Analysis: Example CLI Projects for HelixCode Porting

This document provides a detailed analysis of three AI-powered CLI projects: Qwen Code, Gemini CLI, and DeepSeek CLI. Focus is on features that can be ported to HelixCode, including model-specific implementations and LLM integrations.

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1. QWEN CODE

Overview

Qwen Code is an AI-powered CLI tool adapted from Gemini CLI, specifically optimized for Alibaba's Qwen3-Coder models. It's built with TypeScript/Node.js and designed to handle large codebases with advanced code understanding.

Repository: github.com/QwenLM/qwen-code
Language: TypeScript (Node.js 20+)
Architecture: Monorepo with core, CLI, and test utilities packages

1.1 Core Features & Capabilities

Code Understanding & Editing

- Query and edit large codebases beyond context window limits
- Context compression for handling extensive file contents
- Session management with configurable token limits
- Token tracking via `/stats` command

Workflow Automation

- Git automation (commit analysis, changelog generation)
- File operations automation
- TODO comment extraction and GitHub issue creation
- Pull request and rebase automation

Vision Model Support (Unique Feature)

- **Auto-detection of images** in user input
- **Intelligent model switching** to vision-capable models
- Three modes:
  - `"once"`: Switch for single query, then revert
  - `"session"`: Switch for entire session
  - `"persist"`: Continue with current model
- Command-line override: `--vlm-switch-mode {mode}`
- YOLO mode auto-switches without prompts

Parser Optimization

- Enhanced parser specifically adapted for Qwen-Coder models
- Better handling of Qwen-specific output formats

1.2 Supported LLM Providers & APIs

Primary Authentication Methods:

1. **Qwen OAuth (Recommended)**
   - 2,000 requests/day (no token counting)
   - 60 requests/minute rate limit
   - Browser-based authentication via qwen.ai
   - Automatic credential management and refresh
   - Command: `/auth` to switch if initialized with OpenAI mode
2. **OpenAI-Compatible API**
   - Supports any OpenAI-compatible endpoint
   - Environment variable configuration
   - Project `.env` file support

Regional Provider Options:

For Mainland China: - Alibaba Cloud Bailian: `https://dashscope.aliyuncs.com/compatible-mode/v1` - ModelScope (Free): `https://api-inference.modelscope.cn/v1` - 2,000 free API calls/day - Requires Aliyun account connection

For International Users: - Alibaba Cloud ModelStudio: `https://dashscope-intl.aliyuncs.com/compatible-mode/v1` - OpenRouter: `https://openrouter.ai/api/v1` - Free tier available

1.3 Supported Models

Native Qwen Models: - `qwen3-coder-plus` (Latest mainline coding model) - `qwen3-vl-plus` (Vision model for image understanding) - `qwen3-coder-30BA3B` (Alternative variant) - `qwen3-coder-480B-A35B-Instruct` (Larger variant)

Via OpenAI-Compatible Interface: - Any model available through configured API endpoints

Vision Model Selection:

```
// availableModels.ts shows model metadata structure
const MAINLINE_VLM = 'vision-model';
const MAINLINE_CODER = 'qwen3-coder-plus';

// Models can be marked with isVision flag
const model = { id, label, description, isVision: true };
```

1.4 Key Technical Implementations

Provider Implementation: DashScope (Alibaba)

File: `packages/core/src/core/openaiContentGenerator/provider/dashscope.ts`

Key Features: - Extends OpenAI SDK for compatibility - DashScope-specific caching via headers: - `X-DashScope-CacheControl`: Enable token caching - `X-DashScope-AuthType`: Authentication type - `X-DashScope-UserAgent`: CLI version tracking

Request Building:

```
buildRequest(request: ChatCompletionCreateParams, userPromptId:
        string) {
  // Apply cache control (ephemeral tokens)
  // Add DashScope request metadata (sessionId, promptId)
  // Apply output token limits per model
  // Enable high-resolution images for vision models
  return enhanced_request;
}
```

Cache Control Strategy: - System message + last tool message + latest history message - Ephemeral cache type for non-persistent caching - Automatic message normalization to array format - Token limit enforcement based on model capabilities

Vision Model Detection:

```
isVisionModel(model: string): boolean {
  // Matches: 'vision-model', 'qwen-vl*', 'qwen3-vl-plus*'
  // Special handling for vision-specific parameters
  // vl_high_resolution_images flag
}
```

Authentication Architecture

OAuth Flow: - Browser-based login via qwen.ai - Automatic token refresh - Credential storage and management - Session-based rate limiting

OpenAI-Compatible Auth: - API key environment variable: `OPENAI_API_KEY` - Custom base URL: `OPENAI_BASE_URL` - Custom model selection: `OPENAI_MODEL`

Session Management

Configuration in `~/.qwen/settings.json`:

```
{
  "sessionTokenLimit": 32000,
  "experimental": {
    "vlmSwitchMode": "once",
    "visionModelPreview": true,
    "disableCacheControl": false
  }
}
```

1.5 Unique Features for Porting

1. **Vision Model Auto-Switching**
   - Detects images in prompts automatically
   - User-configurable switch behavior
   - No breaking changes to core API
2. **DashScope Caching System**
   - Ephemeral token caching
   - Session and prompt ID tracking
   - Metadata-driven optimization
3. **Parser-Level Adaptations**
   - Qwen-specific output format handling
   - Better multimodal support
   - Context compression optimizations
4. **Multi-Provider Fallback**
   - OAuth primary, with API key fallback

    - Regional endpoint selection
    - Automatic provider detection
5. **Enhanced Token Management**
    - Session token limits
    - Compression commands (`/compress`)
    - Token usage visibility (`/stats`)

1.6 Commands & Session Features

```
/help       - Display available commands
/clear      - Clear conversation history
/compress   - Compress history to save tokens
/stats      - Show current session information
/exit       - Exit Qwen Code
/auth       - Switch authentication method (OAuth ↔ API key)
```

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2. GEMINI CLI

Overview

Gemini CLI is Google's official open-source AI agent for the terminal, providing access to Gemini models with built-in tools for file operations, shell commands, web search, and MCP (Model Context Protocol) integration.

Repository: github.com/google-gemini/gemini-cli
Language: TypeScript (Node.js 20+)
Architecture: Monorepo with core, CLI, a2a-server, and test utils **License**: Apache 2.0

2.1 Core Features & Capabilities

Code Understanding & Generation

- Query and edit large codebases
- Generate new apps from PDFs, images, or sketches (multimodal)
- Debug and troubleshoot with natural language
- Repository-level analysis and understanding

Automation & Integration

- Operational task automation (PR queries, rebases)
- MCP (Model Context Protocol) servers for custom capabilities
- Non-interactive mode for scripts

- GitHub Actions integration

Advanced Capabilities

- **Google Search Grounding**: Real-time information via built-in Google Search
- **Conversation Checkpointing**: Save and resume complex sessions
- **Custom Context Files**: GEMINI.md for project-specific behavior
- **Token Caching**: Optimize API usage and costs

GitHub Integration

- Pull request reviews with contextual feedback
- Issue triage and automated labeling
- On-demand assistance via `@gemini-cli` mentions
- Custom workflow automation

Built-in Tools

- **File System**: Read, write, create, delete operations
- **Shell Commands**: Execute and analyze system commands
- **Web Fetch & Search**: Retrieve and analyze web content
- **MCP Servers**: Extensible custom integrations

2.2 Supported LLM Providers & APIs

Three Primary Authentication Methods:

1. **Google OAuth (Personal/Code Assist License)**
    - Best for: Individual developers and organization users
    - Free tier: 60 requests/min, 1,000 requests/day
    - Gemini 2.5 Pro with 1M token context window
    - No API key management
    - Command:

      ```
      gemini  # Opens browser for authentication
      ```

    - For Code Assist License (Enterprise):

      ```
      export GOOGLE_CLOUD_PROJECT="YOUR_PROJECT_ID"
      gemini
      ```

2. **Gemini API Key**
    - Best for: Model control and paid tier access
    - Free tier: 100 requests/day
    - Usage-based billing
    - Source: https://aistudio.google.com/apikey
    - Command:

```
export GEMINI_API_KEY="YOUR_API_KEY"
gemini
```

3. **Vertex AI (Enterprise)**
    - Best for: Enterprise teams and production workloads
    - Advanced security and compliance
    - Scalable with higher rate limits
    - Billing account integration
    - Commands:

```
export GOOGLE_API_KEY="YOUR_API_KEY"
export GOOGLE_GENAI_USE_VERTEXAI=true
gemini
```

2.3 Supported Models

Gemini Model Family: - `gemini-2.5-pro` (Default, most capable) - `gemini-2.5-flash` (Fast, cost-effective) - `gemini-2.5-flash-lite` (Lightweight) - `auto` (Automatic model selection)

Advanced Features by Model: - Thinking support: Gemini 2.5 models (default enabled) - Thinking token limit: Capped at 8,192 to prevent loops - Lite models: Excluded from fallback during outages

2.4 Key Technical Implementations

GeminiClient Architecture

- Turn management (max 100 turns per session)
- Loop detection to prevent infinite conversations
- Chat compression for context optimization

- IDE context tracking for multimodal capabilities
- Session telemetry tracking

Thinking Mode

- Supported in Gemini 2.5 models
- Token limit: 8,192 (prevents runaway thinking)
- Disabled by default for flash-lite to save tokens

Model Fallback Strategy

When Gemini Pro is degraded: - Automatically downgrade to Flash - Preserve lite model selections to save costs - Transparent to end user

2.5 Unique Features for Porting

1. **Google Search Grounding**
    - Real-time information integration
    - Search-aware responses
    - Production-ready implementation
2. **Checkpointing System**
    - Save conversation state at specific points
    - Resume from checkpoints
    - Work preservation across sessions
3. **IDE Companion**
    - VS Code integration
    - Context from active editor
    - Multi-file understanding
4. **MCP Protocol Support**
    - Multiple transport modes
    - Custom tool integration
    - Extensible architecture
5. **Loop Detection Service**
    - Prevents infinite agent loops
    - Detects conversation patterns
    - Automatic recovery
6. **Chat Compression Service**
    - Summarization of long conversations
    - Token optimization
    - Configurable compression triggers
7. **Advanced Tools API**
    - Standardized tool interface
    - Tool validation
    - Error handling

2.6 Release Management

Three-tier Release Strategy:

1. **Nightly** (Daily)
    - All changes from main branch
    - Tag: `nightly`
2. **Preview** (Weekly Tuesday, 23:59 UTC)
    - Vetted nightly build
    - Tag: `preview`
    - Full promotion next Tuesday
3. **Stable** (Weekly Tuesday, 20:00 UTC)
    - Full promotion of previous week's preview
    - All bug fixes and validations
    - Tag: `latest`

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3. DEEPSEEK CLI

Overview

DeepSeek CLI is a lightweight, focused command-line AI assistant leveraging DeepSeek Coder models. It emphasizes local-first development with Ollama, plus cloud API options. Minimal dependencies, simple architecture, good for understanding baseline implementation patterns.

Repository: github.com/holasoymalva/deepseek-cli
Language: TypeScript (Node.js 18+)
Architecture: Single-package simple structure **License**: MIT

3.1 Core Features & Capabilities

Code Completion & Generation

- Intelligent code suggestions across 100+ languages
- Complete function generation
- Context-aware completions

Repository-Level Understanding

- Analyze large codebases
- DeepSeek's advanced code comprehension
- Language-agnostic analysis

Code Refactoring & Migration

- Modernize legacy code
- Framework migration assistance
- Best practice implementation

Debugging & Code Review

- Bug identification
- Security issue detection
- Code quality improvements

Project Scaffolding

- New application generation
- Component generation
- Boilerplate creation

Multi-Language Support

100+ languages including: - Python, JavaScript, TypeScript, Java, C++, C#, Go, Rust, PHP, Ruby, Kotlin, Swift, Scala, R, Julia, Dart - Shell, PowerShell, Bash, Dockerfile, Makefile, YAML, JSON, XML - CUDA, Assembly, Verilog, VHDL, Solidity, Protocol Buffer

3.2 Supported LLM Providers & APIs

Two Execution Modes:

1. **Local Mode (Recommended - Default)**
   - Uses Ollama for local model serving
   - Privacy-first approach
   - No API costs
   - Default: `DEEPSEEK_USE_LOCAL=true`
   - Fallback: Auto-enable if no API key configured
2. **Cloud Mode**
   - Direct API calls to DeepSeek platform
   - Requires API key
   - Better for resource-constrained systems
   - Enable: `DEEPSEEK_USE_LOCAL=false`
   - Source: https://platform.deepseek.com/api_keys

3.3 Supported Models

Local Models (via Ollama): - `deepseek-coder:1.3b` (1GB, 2GB RAM) - Lightweight, quick - `deepseek-coder:6.7b` (4GB, 8GB RAM) - Recommended, balanced - `deepseek-coder:33b` (19GB, 32GB RAM) - Largest, most capable

Cloud API: - Same model family available via DeepSeek API

3.4 Key Technical Implementations

API Client

Dual-Mode Architecture: - Local mode: HTTP POST to Ollama at localhost:11434 - Cloud mode: HTTP POST to https://api.deepseek.com/chat/completions

Request Format (Both Modes):

```
{
  "model": "deepseek-coder:6.7b",
  "messages": [
    {
      "role": "system",
      "content": "You are DeepSeek Coder, an AI programming
        assistant..."
    },
    {
      "role": "user",
      "content": "user prompt"
    }
  ]
}
```

Parameters by Mode: - Local: `temperature: 0.1, num_predict: 4096, timeout: 120s` - Cloud: `temperature: 0.1, max_tokens: 4096, timeout: 30s`

CLI Structure

Commands: - `deepseek`: Interactive REPL mode - `deepseek chat "prompt"`: Single prompt mode - `deepseek setup`: Local Ollama environment setup

Interactive Command Features

- Readline-based REPL
- Ollama connection check on startup

- Model availability verification
- Response formatting:
    - Code block syntax highlighting
    - Inline code formatting
    - Header styling
    - Bold text support
- Spinner for loading states
- Command history navigation

Setup Command

- Ollama installation check
- Service startup assistance
- Model download and verification
- Configuration validation

3.5 Dependencies

Minimal Footprint: - axios: HTTP client - chalk: Terminal colors - commander: CLI framework - dotenv: .env parsing - ora: Loading spinner

Node Requirements: 18.0.0+

3.6 Unique Features for Porting

1. **Dual Local/Cloud Architecture**
    - Seamless switching between modes
    - Configuration-driven behavior
    - Graceful fallback logic
2. **Ollama Integration**
    - Direct HTTP API usage
    - Model management commands
    - Health checking
3. **Minimal Dependencies**
    - Only 5 core dependencies
    - Easy to audit and modify
    - Low deployment footprint
4. **Setup Automation**
    - One-command environment setup
    - Dependency installation helpers
    - Verification commands
5. **Response Formatting**
    - Syntax-aware code block highlighting
    - Markdown-style formatting
    - Terminal-optimized display

COMPARATIVE ANALYSIS

Project Comparison Table

Feature	Qwen Code	Gemini CLI	DeepSeek CLI
Language	TypeScript	TypeScript	TypeScript
Architecture	Monorepo	Monorepo	Single package
Auth Methods	OAuth + API Key	OAuth + API Key + Vertex AI	Local (Ollama) + Cloud API
Vision Support	Yes (multimodal)	Implicit (in Gemini 2.5)	No
MCP Support	Yes	Yes	No
Checkpointing	Session limits	Full checkpointing	Not implemented
Search Integration	Provider-specific	Google Search (built-in)	No
Code Complexity	High	High	Low
Dependencies	Many	Many	5 core
Target Use	Enterprise, Chinese regions	Enterprise, all regions	Local-first development
Model Focus	Qwen models	Gemini models	DeepSeek models

Feature Porting Priorities for HelixCode

High Priority (Core Value): 1. Vision model auto-switching (Qwen) - Enables multimodal understanding 2. Google Search grounding (Gemini) - Real-time information 3. MCP Protocol support (Qwen/Gemini) - Custom integrations 4. Token caching strategies (Qwen/Gemini) - Cost optimization 5. Checkpointing system (Gemini) - Long-running task preservation

Medium Priority (Operational): 1. Dual local/cloud architecture (DeepSeek) - Deployment flexibility 2. Model fallback strategies (Gemini) - Reliability 3. Loop detection (Gemini) - Safety 4. Response formatting (DeepSeek) - UX quality 5. Session token limits (Qwen) - Cost control

Lower Priority (Enhancement): 1. Release cadence automation (Gemini) - CI/CD 2. Telemetry integration (Gemini/Qwen) - Analytics 3. IDE companion integration

(Gemini) - VS Code support 4. GitHub Actions integration (Gemini) - Workflow automation

Model-Specific Integrations

Qwen Integration Pattern:

```
// Suggested HelixCode implementation
type QwenProvider struct {
  apiKey      string
  baseURL     string
  authType    string
  sessionID   string
}

// Features to implement:
// - DashScope cache control headers
// - Vision model detection and switching
// - Token limit enforcement
// - Metadata tracking (sessionId, promptId)
```

Gemini Integration Pattern:

```
// Suggested HelixCode implementation
type GeminiProvider struct {
  apiKey              string
  projectID           string  // For Vertex AI
  useVertexAI         bool
  thinkingMode        int
  fallbackEnabled     bool
}

// Features to implement:
// - Thinking mode with token limits
// - Model fallback logic
// - Search grounding configuration
// - Advanced safety settings
```

DeepSeek Integration Pattern:

```
// Suggested HelixCode implementation
type DeepSeekProvider struct {
  useLocal      bool
  apiKey        string
  ollamaHost    string
  modelName     string
```

```
}

// Features to implement:
// - Ollama health checks
// - Local model detection
// - Seamless cloud fallback
// - Setup automation
```

IMPLEMENTATION ROADMAP FOR HELIXCODE

Phase 1: Foundation (Core LLM Integration)

- [ ] Implement DeepSeek provider (simplest, good baseline)
- [ ] Add Qwen OAuth support
- [ ] Implement Gemini API Key support

Phase 2: Advanced Features

- [ ] Vision model auto-switching (Qwen)
- [ ] Token caching strategies
- [ ] Model fallback logic

Phase 3: Integration & Tools

- [ ] MCP protocol support
- [ ] Google Search grounding
- [ ] Checkpointing system

Phase 4: Enterprise Features

- [ ] Loop detection service
- [ ] Chat compression

- [ ] IDE context tracking

Phase 5: Deployment

- [ ] Release automation
- [ ] Telemetry integration
- [ ] GitHub Actions workflows

KEY LEARNINGS

1. **Architecture Choice**: Monorepo pattern allows modular development and isolated testing. Single-package (DeepSeek) is simpler but limits feature separation.
2. **Auth Patterns**: OAuth for frictionless onboarding, API keys for control. Always provide fallback options.
3. **Provider Abstraction**: Use interface-based design to support multiple providers without coupling.
4. **Error Handling**: Model-specific error messages improve UX. Include setup diagnostics.
5. **Configuration**: Environment variables + JSON config files + CLI flags for maximum flexibility.
6. **Vision Support**: Auto-detection of images + intelligent model switching provides seamless multimodal experience.
7. **Reliability**: Fallback strategies, loop detection, and compression systems ensure robust production use.
8. **Cost Optimization**: Token caching, session limits, and compression are critical for cost-effective operations.
9. **Extensibility**: MCP protocol creates plugin ecosystem without core modifications.
10. **Local-First**: Supporting local models (Ollama) alongside cloud APIs appeals to privacy-conscious users.